3 Compton (1923) demonstrated that light behaves like a stream of discrete particles that carry both energy and momentum by scattering X-rays off a light element.

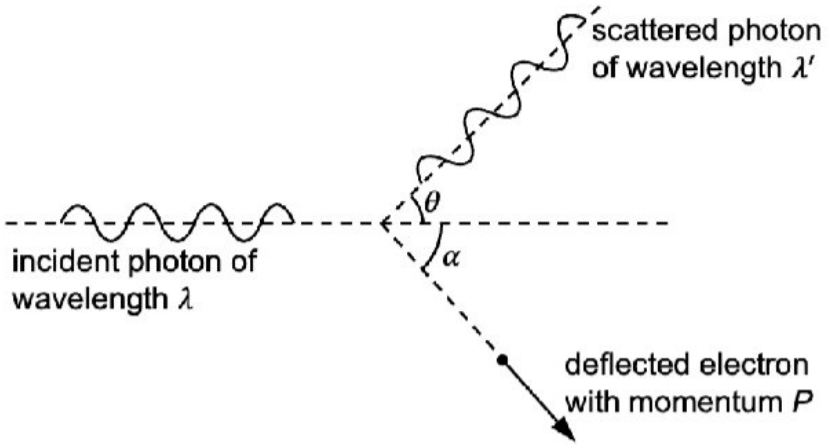


Fig. 3.1

By assuming energy and momentum are conserved in an elastic collision between an incident X-ray photon of wavelength λ and a stationary "free" electron (Fig. 3.1), he derived the equation for the scattered X-ray photon of wavelength λ':

$$\lambda' - \lambda = \frac{h}{mc}(1 - cos\ cos\ \theta)$$

where m is the rest mass of the electron, c is the speed of light in free space and θ is the angle of scattering of the photon.

In one experiment, the incident photon has a wavelength of 191.9×10^{-12} m and θ is 64.3°.

(a) Show that the wavelength of the scattered photon is 193.3×10^{-12} m.

[1]

(b) By using conservation of energy, show that the momentum P of the deflected electron is 3.698×10^{-24} N s.

[3]

(c) Momentum is a vector quantity. By taking components of momentum at right-angles to the direction of the incident photon, calculate the angle α of deflection of the electron as shown in Fig. 3.1.

Explain your working.

α = ... ° [4]

[Total: 8]